DATA 202 — Quiz 4 Retake: Week 5

Retaking SLOs 05A–05C: Clustering & PCA

Name: _____ Date: _____

The Dataset: Campus Library Readers

The campus library (simulated data, as in Quiz 4) built `readers`: one row per student who used the library this fall (260 students), with four habits:

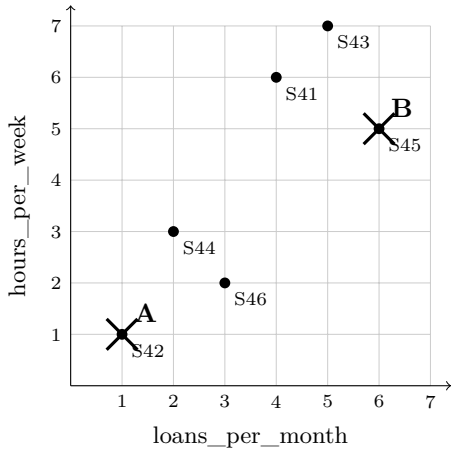
- `loans_per_month` — how many books they borrow per month
- `avg_pages` — the average length of the books they borrow (pages)
- `hours_per_week` — hours they spend in the library per week
- `late_pct` — the % of their loans returned late

A few rows of `readers`:

student_id	loans_per_month	avg_pages	hours_per_week	late_pct
S005	10.6	420	2.8	8
S006	2.1	240	11.4	1
S007	3.3	250	12.1	5
S008	4.5	410	2.6	8

Question 1 (SLO 05A) — Retake from Quiz 3

Run **one round of k-means by hand** (k = 2) on these six students. The starting centroids are **A** = student S42 and **B** = student S45 (the crosses). First write the closest centroid for each student, then compute the two new centroids.



`student`	loans	hours	Closest centroid (A/B)
`S41`	4	6	
`S42`	1	1	
`S43`	5	7	
`S44`	2	3	
`S45`	6	5	
`S46`	3	2	

New centroid A = (_____ , _____) New centroid B = (_____ , _____)

Question 2 (SLO 05B) — Retake from Quiz 3

The library scaled the four columns of `readers` into `X_scaled` and ran k-means on it for k = 1 to 8, recording the inertia and the silhouette score of each run. **Which k would you choose?** Justify using both rows of the table.

k	1	2	3	4	5	6	7	8
inertia	1040	658	379	185	164	151	139	126
silhouette	—	0.38	0.49	0.59	0.49	0.45	0.35	0.26

Question 3 (SLO 05C) — Retake from Quiz 3

The library ran PCA on the same four scaled columns (`X_scaled`):

```
PCA().fit(X_scaled).explained_variance_ratio_.round(3)
# [0.456 0.306 0.195 0.043]
```

How many components would you keep to retain at least **90%** of the variance? What share of the variance do they keep?